Muthu Shravan Sundaram

+31 6 17585933 | shravanmuthu1809@gmail.com | linkedin.com/in/muthu-shravan-sundaram | Portfolio

Professional Summary

MSc graduate in Mechanical Engineering from TU Delft, with a thesis grade of 9/10. Designed and built a modular flexible endoscope from requirements to a tested prototype, integrating a tendon-driven bending section with fluid channels and electronics. Earlier work includes a flexure-based joint for an ESA vibration-isolation system, the mechanical design of underwater vehicles, and CFD simulations of acoustic mixing in microfluidics. Based in the Netherlands on a Zoekjaar permit, free to work without sponsorship until September 2027.

Education

Delft University of Technology — Delft, Netherlands
MSc Mechanical Engineering — *September 2024 – August 2026*

- **Thesis:** A Modular, Affordable, Flexible Endoscope System
- **Thesis Grade: 9/10**

IIITDM Kancheepuram — Chennai, India
BTech Mechanical Engineering with Honors — *July 2020 – June 2024*

- **CGPA: 9.49/10**
- Awarded the **Institute Gold Medal** for the highest CGPA across all disciplines in the 2024 graduating cohort

Technical Skills

Software & CAD: SOLIDWORKS, Fusion 360, COMSOL Multiphysics, AutoCAD, MATLAB, ANSYS
Core Competencies: Mechanical Design, DFM/DFA, FEA, CFD, Structural Analysis, Rapid Prototyping, BOM Management, Technical Drawings, Functional Testing & Validation, DFMEA, Root Cause Analysis, Basic GD&T
Programming: Python, C, C++

Experience

Medical Device Engineering Intern — July 2025 – October 2025
Erasmus MC — *Rotterdam, Netherlands*

- Conducted an observational product journey and workflow analysis of flexible endoscopes at two hospitals, covering clinical use from preparation to post-procedure, reprocessing, maintenance and repair, and end of life.
- Identified damage hotspots, failure modes, and operational and design challenges through observation and discussions with endoscopists, reprocessing staff, repair technicians, and Olympus Uptime Support.
- Documented findings in a product journey and workflow analysis report that informed the requirements of the MSc thesis.

Research Intern — May 2022 – May 2024
Microscale Transport Laboratory, IIITDM Kancheepuram — *Chennai, India*

- Modeled acoustofluidic mixing in COMSOL Multiphysics, coupling Navier-Stokes flow with acoustic pressure fields, and ran parametric studies on frequency, acoustic energy density, and inlet velocity. Co-authored the resulting paper, which won the Best Paper Award at ICRAFMN 2023.
- Fabricated circular well microchannel prototypes and experimentally validated acoustic fluid relocation behaviors.
- Performed hands-on experimental support for acoustic streaming studies in rectangular microchannels, assisting lab researchers with physical flow characterization.

Team Lead & Mechanical Design Member - AUV Society — October 2021 – August 2023
IIITDM Kancheepuram — *Chennai, India*

- Progressed from Mechanical Design Member (SAUVC & MATE 2022) to Team Lead (MATE 2023), managing a 12-person cross-functional team across mechanical, electrical, and software divisions.
- Managed technical team budgets, defined system-level requirements, and oversaw the integration of sub-assemblies.
- Generated 2D manufacturing drawings in Fusion 360 and coordinated with external fabricators, ensuring all custom components met appropriate tolerances and cost constraints.

Projects

Affordable Modular Flexible Endoscope - Master's Thesis — February 2026 – August 2026

- Designed and developed a modular, flexible endoscope from clinical requirements through CAD, rapid prototyping, and physical validation.
- Engineered the highly constrained internal architecture in SOLIDWORKS, integrating fluidic routing (suction, irrigation, insufflation) and a tendon-driven bending section reaching 210°–214° in all four directions.
- Executed Design Failure Mode and Effects Analysis (DFMEA) across mechanical, electronic, and fluidic subsystems before each build, covering 18 failure modes and implementing a mitigation for each, prioritizing the three high-risk modes in the electronics and tendon channels.
- Designed a toolless, watertight, modular detachable interface between the handle and insertion tube through iterative FDM prototyping and testing, sealing the fluid channels with tapered dispensing needle tips that reseal on every reconnection.

Compliant Planar Joint for ESA Microvibration Isolation System April 2025 – June 2025

- Collaborated in a team to develop a flexure-based planar joint for the European Space Agency (ESA) Microvibration Isolation System, using MATLAB-based constrained optimization on folded leaf spring geometries to meet strict 6-DOF stiffness targets.
- Developed the complete parametric CAD assembly in SOLIDWORKS and ran the multi-axis COMSOL FEA with one teammate to validate yield safety, stress distribution, and large-stroke performance.

Mechatronic Assistive Feeding Robot February 2025 – June 2025

- Designed the SOLIDWORKS mechanical architecture of a 3-DOF telescopic robotic arm to help patients with multiple sclerosis feed themselves, and sized the Maxon motors from payload calculations.
- Developed the theoretical control architecture for a dual-stage tremor damping system, formulating PID/notch filter logic block diagrams to enable switching between active feeding assistance and physical rehabilitation modes.

Circular Well Acoustic Micromixer May 2023 – May 2024

- Developed a 2D CFD model in COMSOL Multiphysics to simulate acoustic mixing of inhomogeneous co-flowing fluids in a circular well microchannel using Bessel-function pressure fields.
- Formulated a transient mixing index incorporating instantaneous velocity and concentration to quantify mixing efficiency.
- Showed that alternating the frequency between 1 and 2 MHz raised the mixing index to 0.902 in simulation, against 0.355 with diffusion alone (2.5 times higher).

Subsea Robotics: AUV & ROV Development October 2021 – October 2022

- Collaborated on the mechanical design of a 5-DOF, battery-powered Autonomous Underwater Vehicle (AUV) for SAUVC 2022 with dual waterproof pressure hulls and four vectored thrusters, designing the custom aluminum frames and clamps.
- Optimized an existing Remotely Operated Vehicle (ROV) frame for stiffness with slight positive buoyancy as a fail-safe, and engineered a pneumatic robotic arm with one cylinder driving a two-link elbow and a second actuating the gripper.
- Solved recurring hull leaks at the cable exits by replacing rigid potting with a flexible marine epoxy and verified the seal in progressively longer pool tests before SAUVC 2022 and the MATE ROV World Championship 2022.

Publications & Presentations

Acoustic Mixing of Co-flowing Inhomogeneous Fluids in a Circular Well Microchannel *Best Paper Award*

- Sujith Jayakumar, **Muthu Shravan Sundaram**, Varun Kumar Rajendran, Karthick Subramani
- ICRAFMN 2023, Bangalore, India

Mechanical Design, Analysis and Development of an AUV for SAUVC *Oral Presentation*

- J R Sukesh, **Muthu Shravan Sundaram**, Vishnuvardhan Iyengar
- Singapore Autonomous Underwater Vehicle Challenge (SAUVC) Conference 2022, Singapore

Design and Performance Analysis of a Bio-Inspired Remotely Operated Robot *Published*

- J R Sukesh, R Dasyam, **Muthu Shravan Sundaram**, V Iyengar
- IEEE OCEANS 2022, Chennai, India